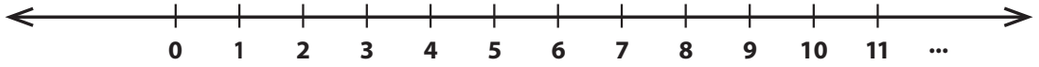
2.2 Introduction of Integers and its representation on a number line

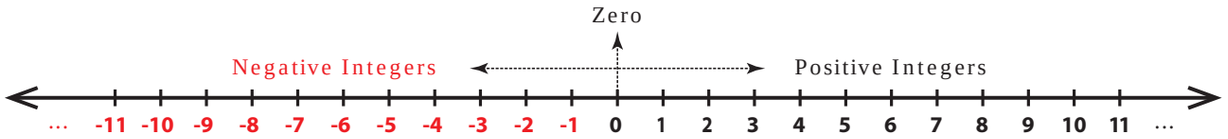
We know that when zero is included to the set of natural numbers then the set of numbers is called as ***Whole numbers***.

Now, let us recall the number line which shows the representation of whole numbers.



Whole numbers

We have seen the need to extend the number line beyond 0 to its left. We call the numbers –1, –2, –3, … (to the left of zero) as negative numbers or negative integers and the numbers 1, 2, 3,… (to the right of zero) as ***positive numbers*** or ***positive integers.*** Hence, the new set of numbers …, -3, -2, -1, 0, 1, 2, 3,… are called ***Integers.*** It is denoted by the letter '**Z**'. The ***Integers*** are shown in the number line below.



The 'plus' and 'minus' sign before a number tells, on which side the number is placed from zero. '–' symbol in front of a number is read as 'negative' or 'minus'. For example, –5 is read as negative 5 or minus 5.

Note

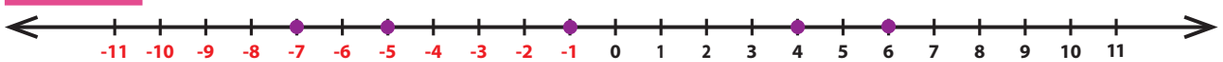
- The number line can be shown both in horizontal and vertical directions.
- The number 0 is neither positive nor negative and hence has no sign.
- Natural numbers are also called as ***positive integers*** and Whole numbers are also called as ***non-negative integers***.
- The positive and the negative numbers together are called as ***Signed numbers***. They are also called as ***Directed numbers***.
- A number without a sign is considered as a positive number.
  For example, 5 is considered as +5.

The letter '**Z**' was first used by the Germans, because the word for Integers in the German language is "**Zahlen**" which means "**number**".

DO YOU KNOW?

Example 1 Draw a number line and mark the integers 6, –5, –1, 4 and –7 on it.

Solution



Try these

1 Read the following numbers orally.
  i) +24 ii) –13 iii) –9 iv) 8
2 Draw a number line and mark the following integers.
  i) 0 ii) –6 iii) 5 iv) –8
3 Are all natural numbers integers?
4 Which part of the integers are not whole numbers?
5 How many units should you move to the left of 3 to reach –4?

E6J8HD

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